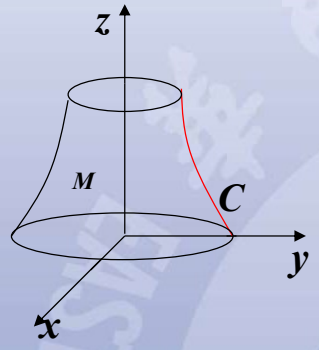
高等数学（下）

理学院　胡海燕

一、特殊曲面

1. 旋转曲面

定义 一条平面曲线绕其平面 上的一条定直线旋转一 周所成的曲面叫做 旋转曲面，这条定直线叫做旋转曲 面的 轴.



考察 yoz 坐标面上一条曲线 C，

其方程为： $f(y,z)=0.$

将曲线 C 绕 z 轴旋转一周就得到 一个以 z 轴为轴的旋转曲面. 求此旋转曲面的方程.

设$M(x,y,z)$ 为曲面上的任一点，

该点必为C上某点$M_1(0,y_1,z_1)$旋转得到

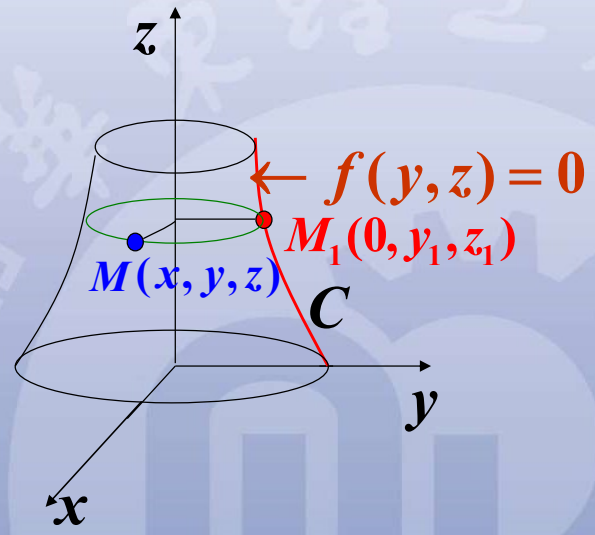
则有 $z=z_1$

且点M、M_1 到z轴的距离相同

$\sqrt{x^2+y^2}=|y_1|$ 即$y_1=\pm\sqrt{x^2+y^2}$

而$f(y_1,z_1)=0$

从而有 $f(\pm\sqrt{x^2+y^2},z)=0.$



2. 圆锥面

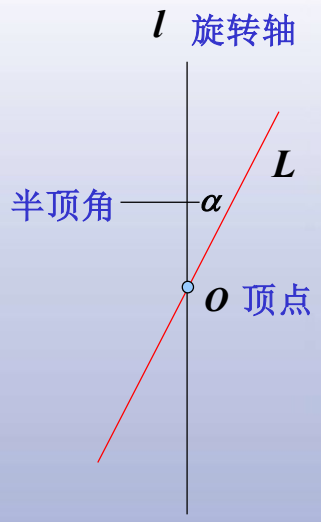
直线 L 绕着另一条与L相交的定直线旋转一周，

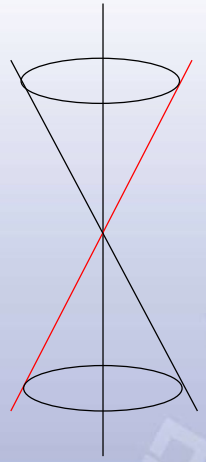
所得的 旋转曲面叫做 圆锥面.

两直线的交点称为圆锥 面的 顶点，

两直线的夹角 α $(0<\alpha<\frac{\pi}{2})$ 称为圆锥面的

半顶角， 与 L 相交 的直线称为 旋转轴.

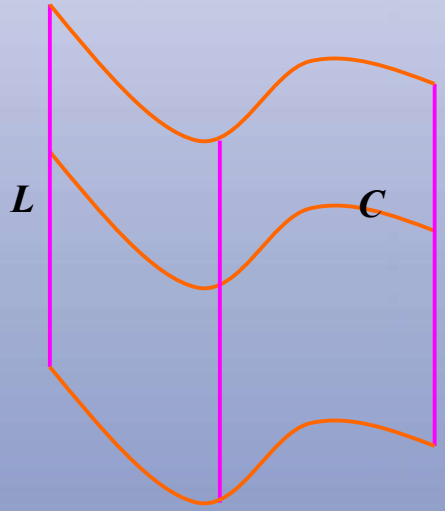




一般方程（绕z轴）： $x^2 + y^2 = k^2 z^2$

3. 柱面

定义 平行于定直线并沿着已 知曲线 C (不与动直线在同一平面 上)移动的直线 L 形成的轨迹叫做柱面， 定曲线 C 叫做柱面的 准线，动直线 L 叫做柱面的母线.



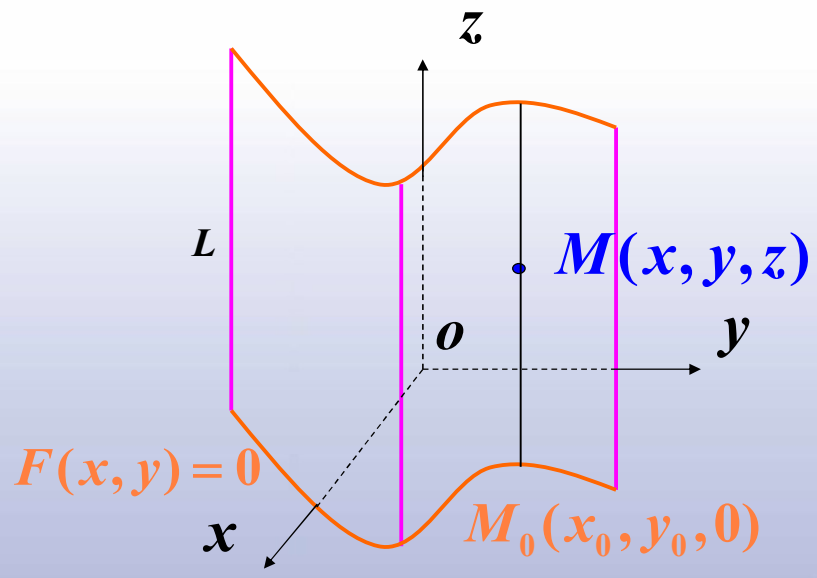
注. 1、我们只讨论母线平行 于坐标轴的柱面 .

2、只含 x, y 而缺 z 的方程 $F(x, y) = 0$,

在空间直角坐标系中表 示

母线平行于 z 轴的柱面,

其准线是 xoy 面上的曲线 C: $F(x, y) = 0$.



$$\begin{cases} x = x_0 \\ y = y_0 \end{cases} \qquad F(x_0, y_0) = 0,$$

$$F(x, y) = 0.$$

二、二次曲面

三元一次方程表示的曲　面称为一次曲面，即平面.

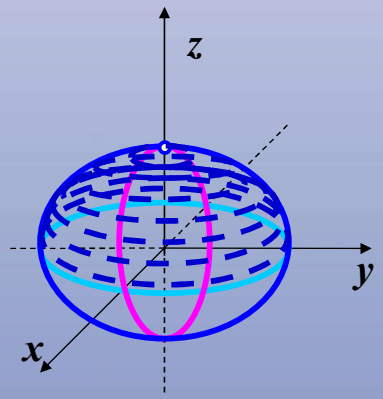
三元二次方程表示的曲 面称为 二次曲面.

研究方法：截痕法（俗称CT扫描法）

1、椭球面

由方程 $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}+\dfrac{z^2}{c^2}=1 \quad (a>0, b>0, c>0)$

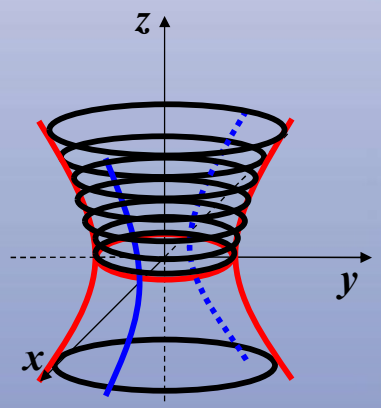
确定的曲面称为 椭球面

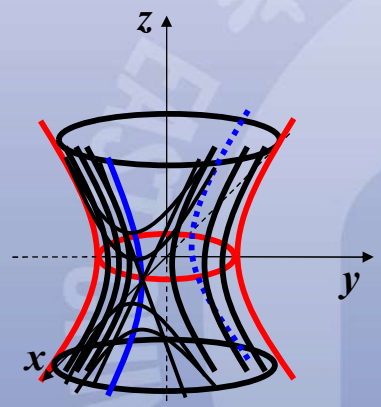


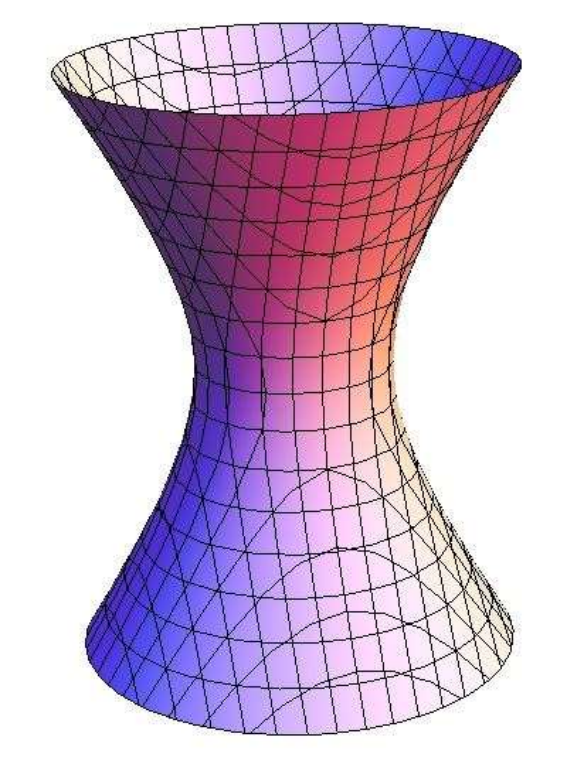
2、单叶双曲面

由方程 $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}-\dfrac{z^2}{c^2}=1 \quad (a>0,\ b>0,\ c>0)$

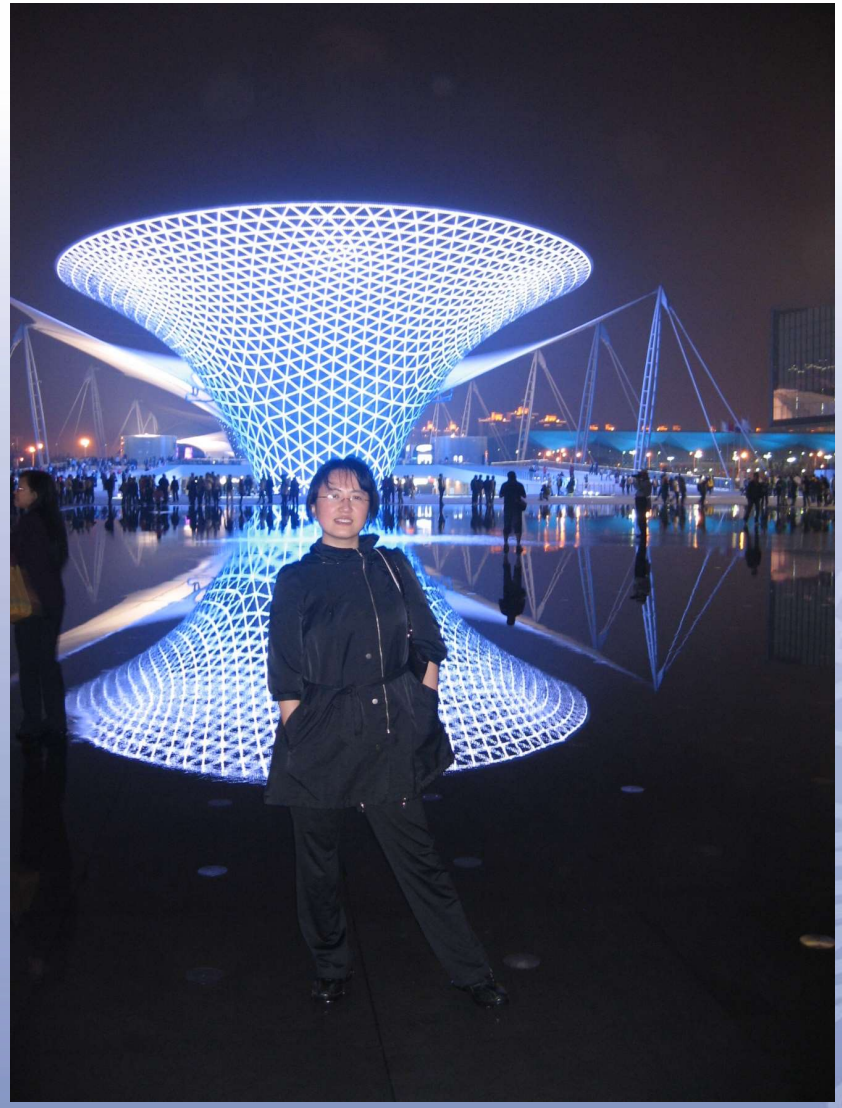
确定的曲面称为 单叶双曲面





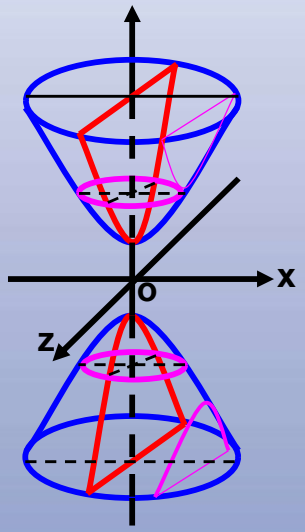


风继续吹



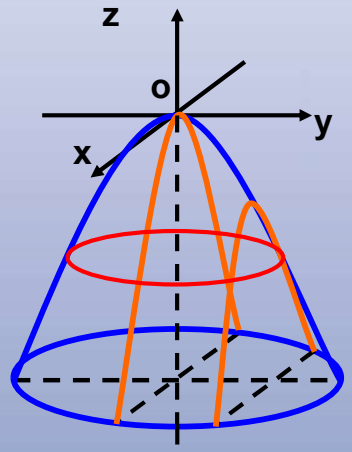
3、双叶双曲面

$$\frac{y^2}{b^2}-\frac{x^2}{a^2}-\frac{z^2}{c^2}=1(a>0,\ b>0,\ c>0)$$

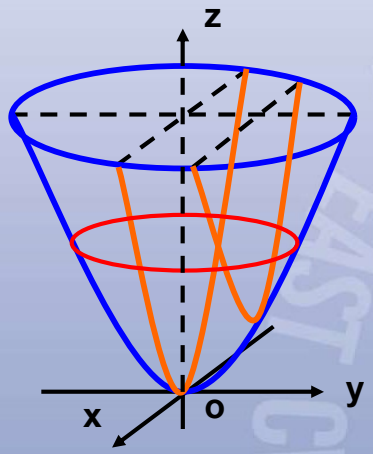


4、椭圆抛物面

$$\frac{x^2}{p}+\frac{y^2}{q}=2z(pq>0)$$



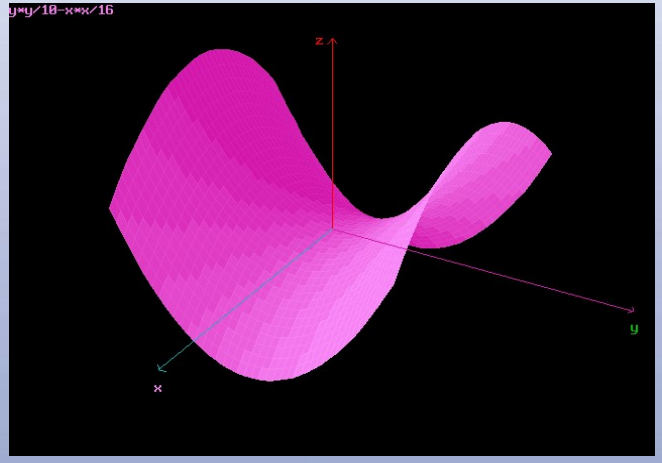
$p<0,\quad q<0$



$p>0,\quad q>0$

5、双曲抛物面（马鞍面）

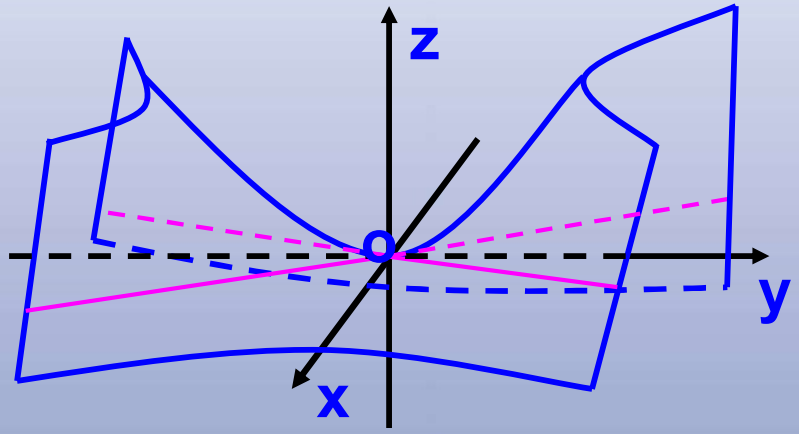
$$-\frac{x^2}{p}+\frac{y^2}{q}=2z(pq>0)$$

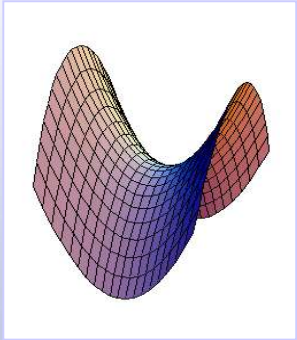


$p>0$

双曲抛物面（马鞍面）

图形如下：

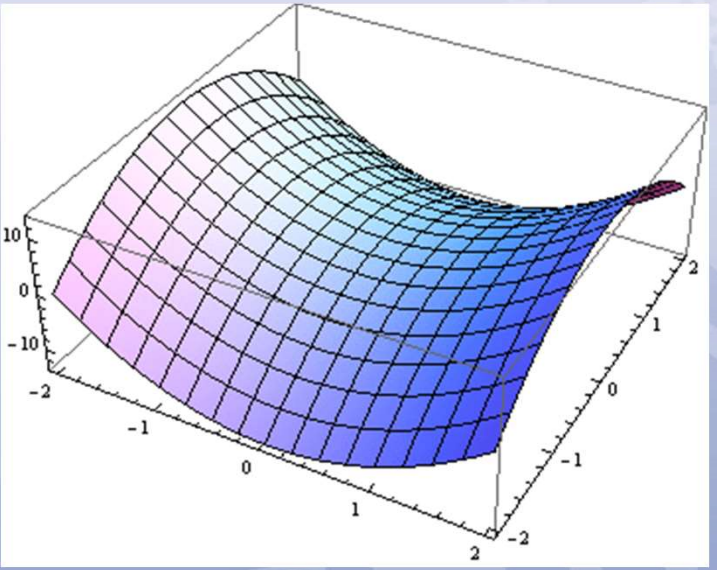




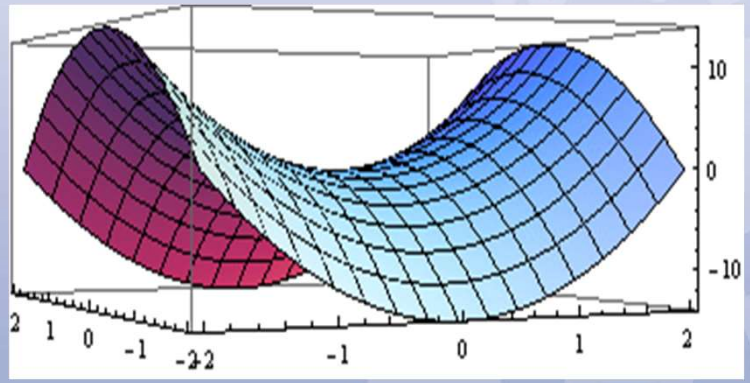
注：1、马鞍面也是直纹面；

2、$z = xy$也是马鞍面。

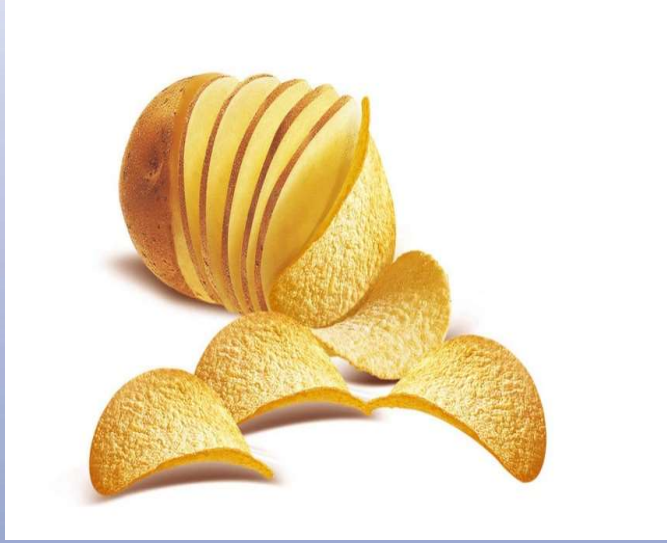
三、双曲抛物面的应用（鸟巢）

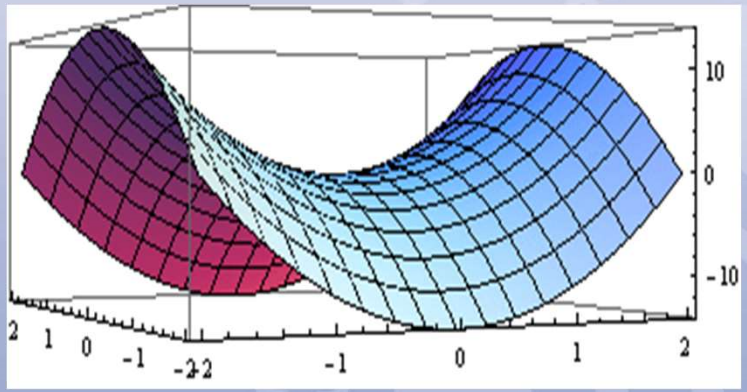


三、双曲抛物面的应用(广州星海音乐厅)



三、双曲抛物面的应用（薯片）

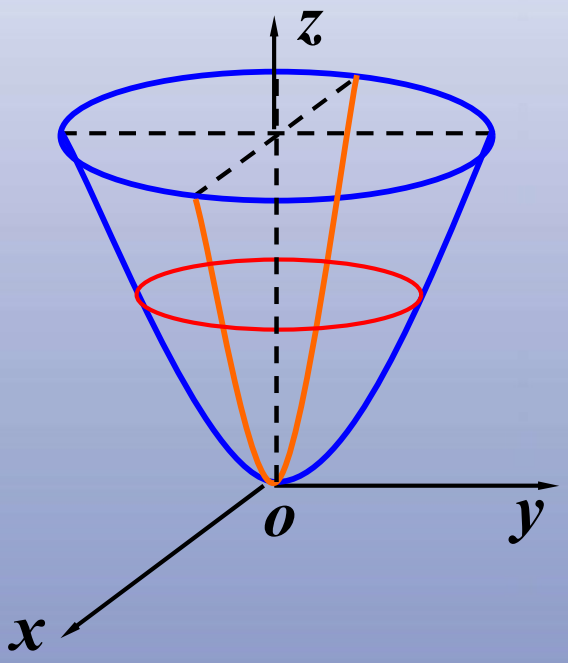


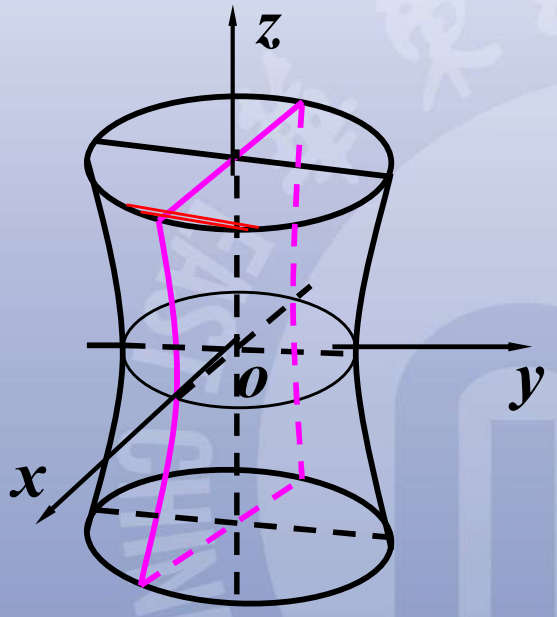


一、画出方程所表示的曲面：

1、$\dfrac{z}{3}=\dfrac{x^2}{4}+\dfrac{y^2}{9}$；

2、$16x^2+4y^2-z^2=64$.

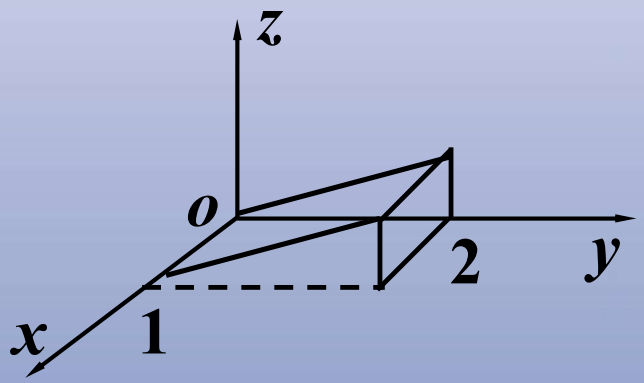


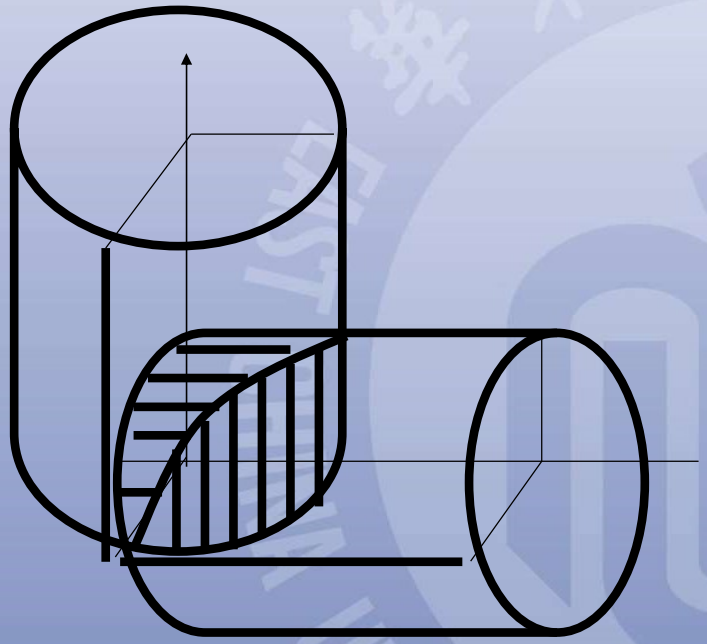


二、 画出下列各曲面所围成的立体的图形：

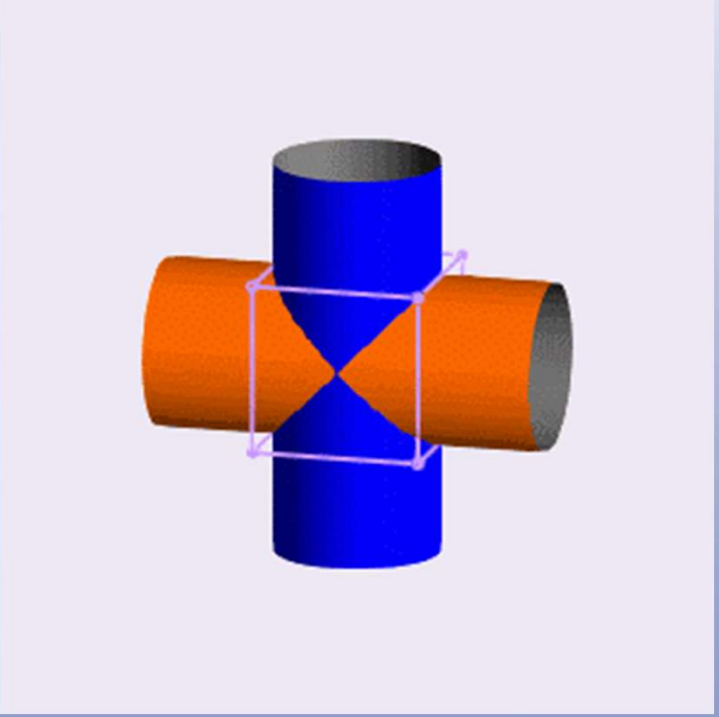
1、$x=0$, $z=0$, $x=1$, $y=2$, $z=\dfrac{y}{4}$;

2、$x=0$, $y=0$, $z=0$, $x^2+y^2=R^2$, $x^2+z^2=R^2$
（在第一卦限内）.



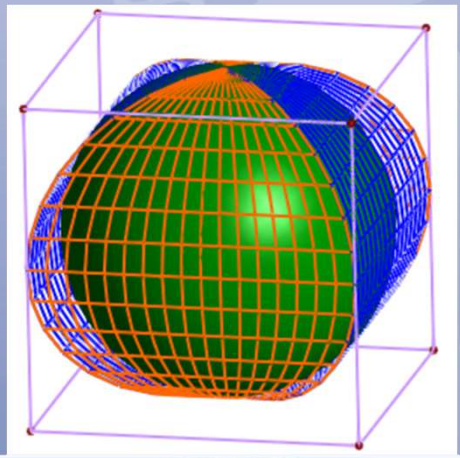


牟合方盖



刘徽

约公元225年—295年



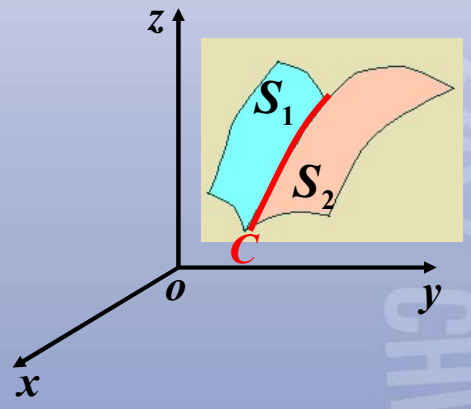
三、空间曲线

1、空间曲线的一般方程

空间曲线C可看作空间两曲面的交线.

$$\begin{cases} F(x,y,z)=0 \\ G(x,y,z)=0 \end{cases}$$

空间曲线的一般方程



特点： 曲线上的点都满足方程，满足方程的点都在曲线上，不在曲线上的点不能同时满足两个方程.

2、空间曲线的参数方程

$$\boxed{\begin{cases} x = x(t) \\ y = y(t) \\ z = z(t) \end{cases}}$$ 空间曲线的参数方程

当给定$t = t_1$时，就得到曲线上的一个点(x_1, y_1, z_1)，随着参数的变化可得到曲线上的全部点.

3、空间曲线在坐标面上的投影

设空间曲线的一般方程：$\begin{cases} F(x,y,z)=0 \\ G(x,y,z)=0 \end{cases}$

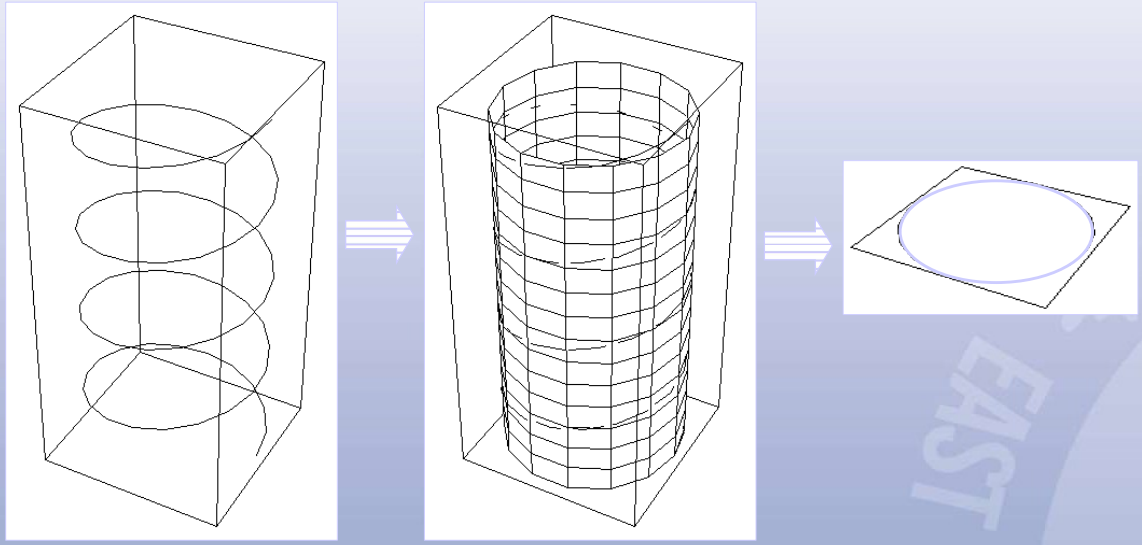
消去变量**Z**后得： $\underline{H(x,y)=0}$

曲线关于xoy 的**投影柱面**

投影柱面的特征：

以此空间曲线为准线，垂直于所投影的坐标面.

如图:投影曲线的研究过程.



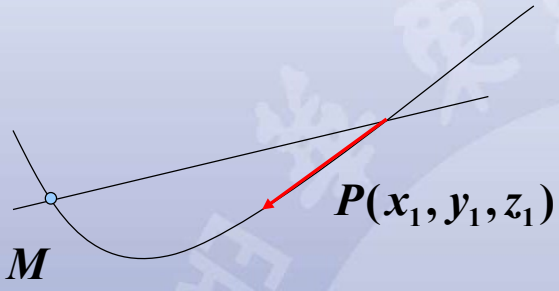
空间曲线　　投影柱面　　投影曲线

4、空间曲线的切线和弧长

设曲线 $\begin{cases} x = x(t) \\ y = y(t) \\ z = z(t) \end{cases}$ 考虑在点 $M(x_0, y_0, z_0)$处的切线 (此时 $t = t_0$),

割线 MP 的方程为：

$$\frac{x - x_0}{x_1 - x_0} = \frac{y - y_0}{y_1 - y_0} = \frac{z - z_0}{z_1 - z_0}$$



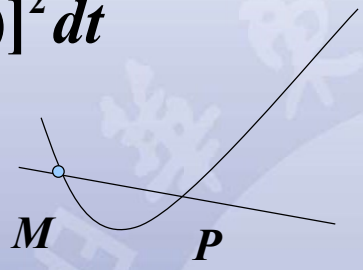
即 $$\frac{x - x_0}{x(t_1) - x(t_0)} = \frac{y - y_0}{y(t_1) - y(t_0)} = \frac{z - z_0}{z(t_1) - z(t_0)}$$

$$\frac{x - x_0}{x(t_0 + \Delta t) - x(t_0)} = \frac{y - y_0}{y(t_0 + \Delta t) - y(t_0)} = \frac{z - z_0}{z(t_0 + \Delta t) - z(t_0)}$$

切线方程为： $$\frac{x - x_0}{x'(t_0)} = \frac{y - y_0}{y'(t_0)} = \frac{z - z_0}{z'(t_0)}$$

$$(ds)^2 \approx |\overrightarrow{MP}|^2 = (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2$$

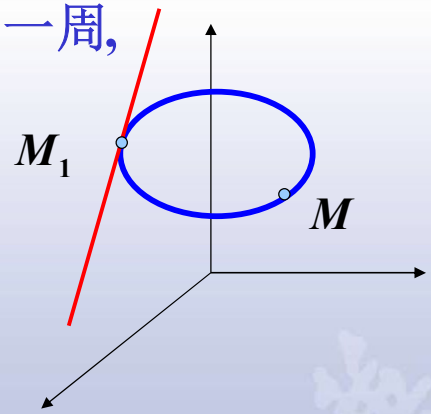
$$ds = \sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2}\,dt$$



弧长公式为：

$$s = \int_a^b \sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2}\,dt$$

例 直线 $L:\dfrac{x-1}{0}=\dfrac{y}{1}=\dfrac{z}{1}$ 绕 z 轴旋转一周，求旋转曲面的方程.



解 设曲面上任意一点 $M(x,y,z)$，

由 $M_1(x_1,y_1,z_1)$

旋转后 到达 $M(x,y,z)$ 位置

由于高度不变，有 $z=z_1$，

又 M 和 M_1 到 z 轴的距离 r 不因旋转而改变，

故 $r^2 = x_1^{\ 2} + y_1^2 \ = x^2 + y^2$,

由于 M_1在直线L上，有$x_1 = 1, y_1 = z_1$

故消去参数后所求旋转曲面方程为

$$x^2 + y^2 - z^2 = 1.$$

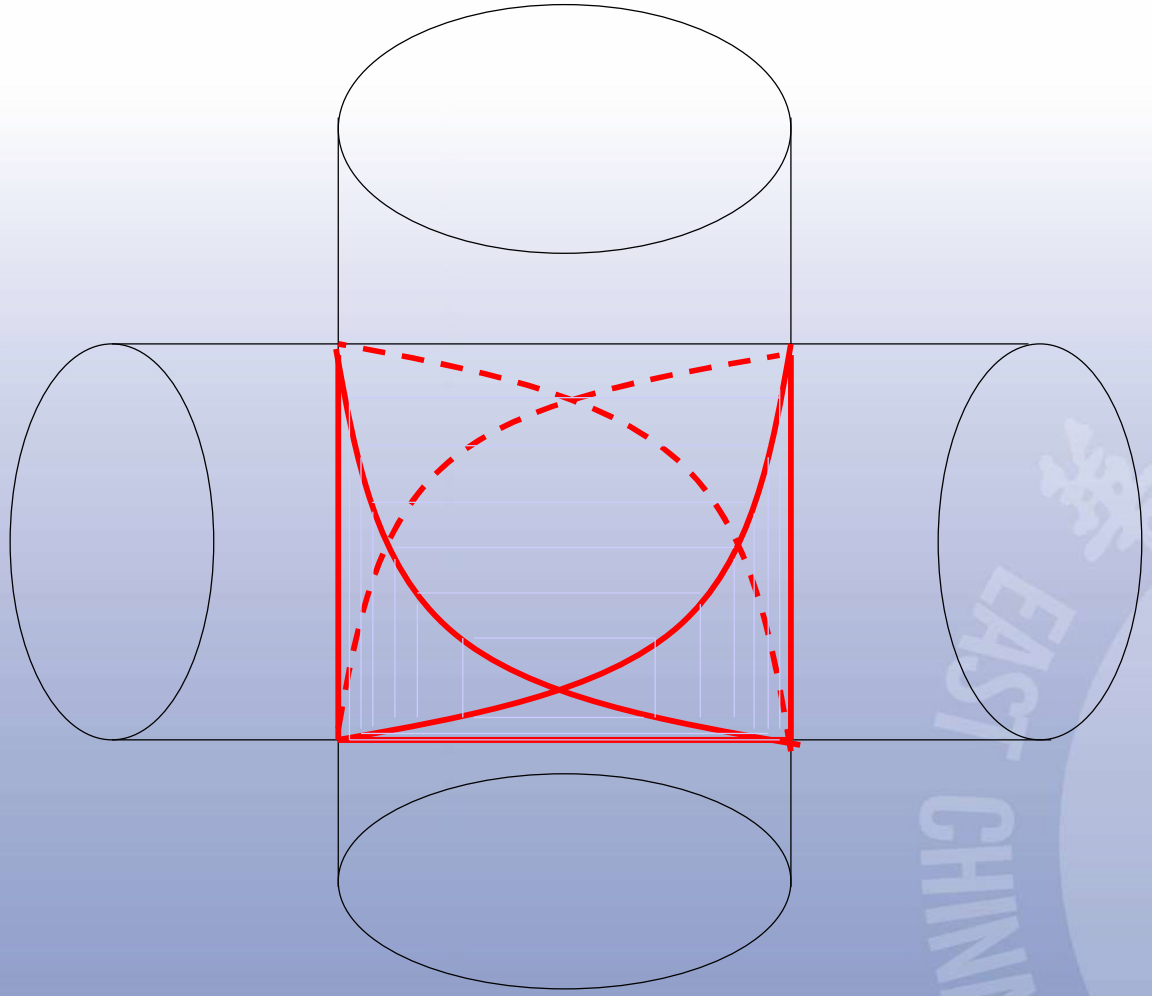
试画出曲线 $\begin{cases} x^2+y^2=1 \\ x^2+z^2=1 \end{cases}$.

解：原方程等价于

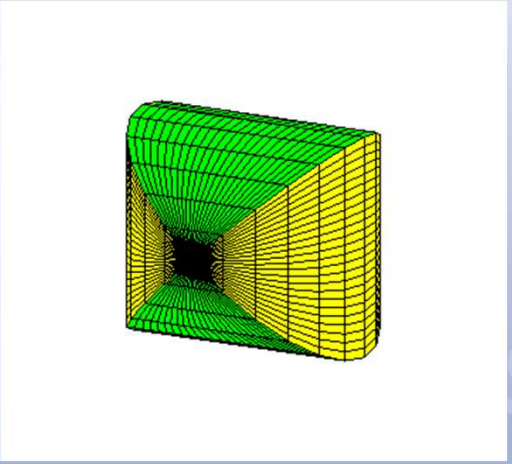
$$\begin{cases} x^2+y^2=1 \\ |y|=|z| \end{cases}$$

即

$$\begin{cases} x^2+y^2=1 \\ y=\pm z \end{cases}$$



牟合方盖



判断下列方程代表什么　图形？

开口向下，椭圆抛物面　$(1)\ z+2x^2+y^2=0$

单叶旋转双曲面　$(2)\ 2x^2+2y^2=1+3z^2$

两个相交平面　$(3)\ x^2-2y^2=0$

锥面　$(4)\ x^2+y^2-(z-1)^2=0$

圆柱面　$(5)\ x^2+y^2=1$

双叶双曲面　$(6)\ x^2+\dfrac{y^2}{2}=\dfrac{z^2}{3}-1$

抛物柱面　$(7)\ z^2=5x$

双曲抛物面　$(8)\ x^2-y^2=4z$

双叶旋转双曲面　$(9)\ x^2-y^2-z^2=4$

Z 轴　$(10)\ x^2+y^2=0$

下半个锥面　$(11)\ z=4-\sqrt{x^2+y^2}$

课后探索下列曲面的形状：

$$xy + yz + zx = 0$$